

A reversible key uses each output letter once. One turn replaces every letter using that key; message positions stay fixed. The next turn uses the new message.

Problem 1: Which key takes longest to bring ABCDE back? Find the first return time of each.

in	A	B	C	D	E
R	B	C	D	E	A

in	A	B	C	D	E
S	B	C	A	E	D

in	A	B	C	D	E
T	B	C	D	A	E

in	A	B	C	D	E
U	B	A	D	C	E

Problem 2: Find every possible first return time for ABCDE. Make a key for each, and show that no other time is possible.

Problem 3: Start with ABCDE. Try each pair of keys in both orders. When does reversing the order change the result? Find a rule.

in	A	B	C	D	E
P	B	A	C	D	E
Q	A	C	B	D	E
L	A	B	C	E	D
V	B	C	A	D	E
W	C	A	B	D	E

P and Q

P and L

V and W

Problem 4: One machine turn is P followed by Q. How many turns bring ABCDE back? Make a machine from P and Q that undoes one turn.